

Assignment 5 - RGB Sensor

Description:

This project implements an RGB color detection system using the TCS34725 RGB sensor with a Raspberry Pi. The system is designed as a reusable library/module that can be easily integrated into other projects. The library reads color data from the sensor using I2C communication and converts the raw values into RGB format. The main program uses this library to output both the detected color in hexadecimal format and the closest matching color name along with a confidence value. The system detects colors based on a predefined palette of 16 colors and provides consistent and readable results

Approach / What was my plan:

First, we will connect the TCS34725 RGB sensor to the Raspberry Pi using the I2C interface and verify communication using system tools. Then, we will develop a hardware interface layer to handle I2C read and write operations using the bcm2835 library.

Next, we will design a custom color detection library that reads raw RGB values from the sensor, normalizes them, and converts them into hexadecimal RGB888 format. We will define a palette of 16 standard colors:

- choosing 16 colors:
 - red
 - burgundy
 - orange
 - brown
 - yellow
 - green
 - lime
 - olive
 - blue
 - purple
 - magenta
 - lavender
 - pink
 - gray
 - white
 - black
- How we are going to identify it: euclidean distance between two color hex codes
- get confidence value by $1 - \text{distance} / \text{total distance} * 100$
- Deciding to use 16-bit color RGB565

Eric Ahsue: Jasuv
Kiran Khatri: khatri5034
Haibin Cao: haibinc
John Tsiglieris: John-Tsiglieris
CSC615 Embedded Linux

ID: 922711514
ID: 923750019
ID: 923756077
ID: 923593954

Issues and Resolutions:

- issues with fine tuning the color identification
 - switch reading from RGB565 to RGB888 for more accurate readings
 - switching out some colors from our original ones (i.e maroon -> burgundy)
 - picking the right hex codes for the best identification of colors
 - messing around with light filters (gain, integration time, etc.)
- adjusting the identification algorithm
 - Euclidian distance formula?
 - Predefined Color Hex ranges?
 - Confidence levels should be more well defined
 - switched to using a range of predefined research values

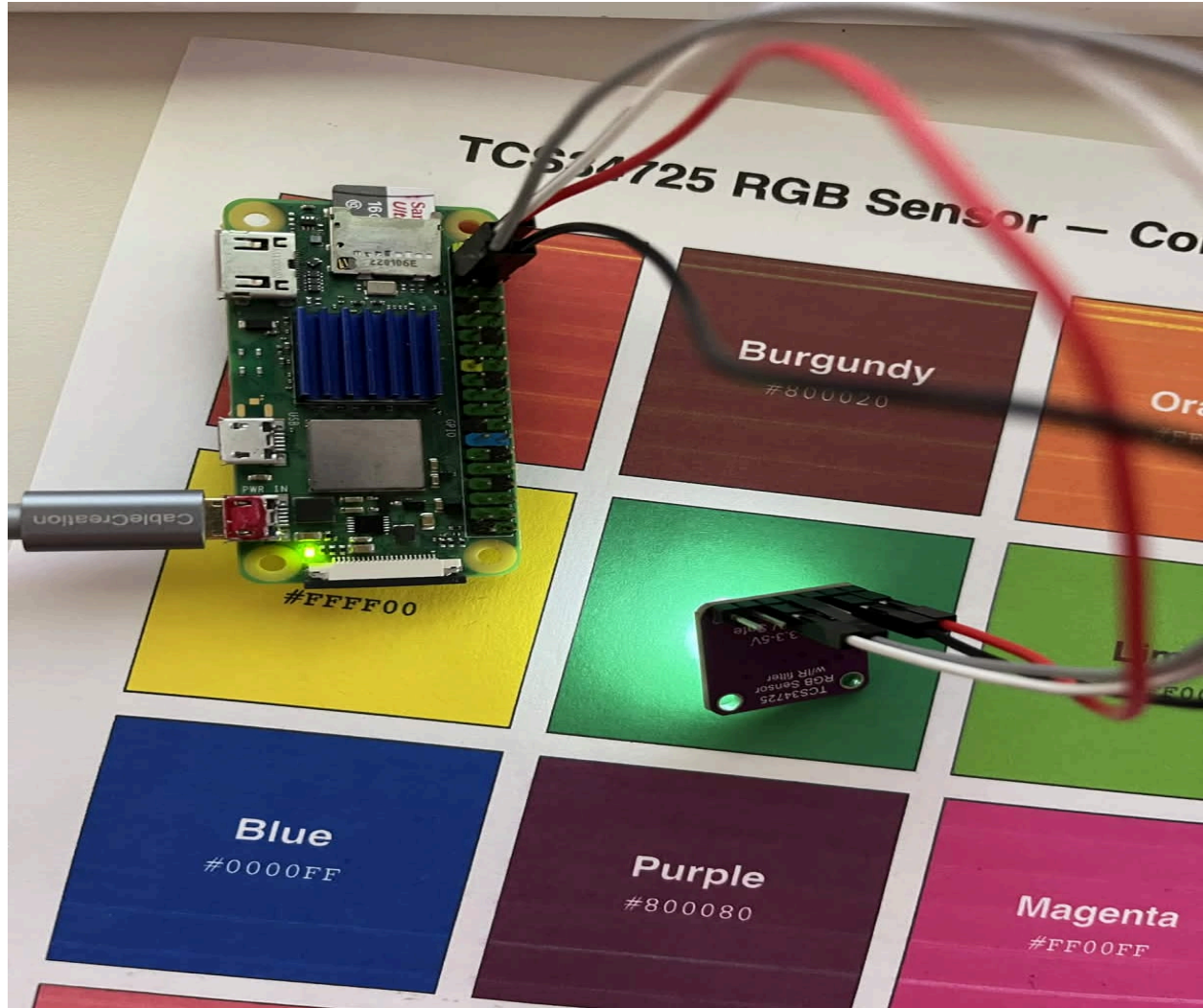
Analysis:

Use Section: Include all Color_Lib and Dev_Config files. Dev_Config handles the I2C communication using the bcm2835 library and Color_Lib handles the register definitions and RGB color values. The bcm2835 library must also be installed on the system. In your function, initialize the Dev_Module with DEV_ModuleInit() and initialize the Color Lib with ColorLib_Init(). Color Lib initializes the I2C target address to 0x29 and reads the device ID to confirm your sensor is connected, sets your integration time and gain, and enables the sensor. Call ColorLib_GetMatch() to get the color match that the sensor returns, every call to ColorLib_GetMatch() returns a reading. You can modify the color ranges and predefined RGB color values within PROJECT_COLORS[]. The first argument in PROJECT_COLORS[] is the color in english, second argument is the color in hexadecimal value, the third and fourth argument are the red minimum value and red maximum values (0-255), fifth and sixth are the green minimum value and green maximum values (0-255), and seventh and eighth arguments are the blue minimum and blue maximum values (0-255).

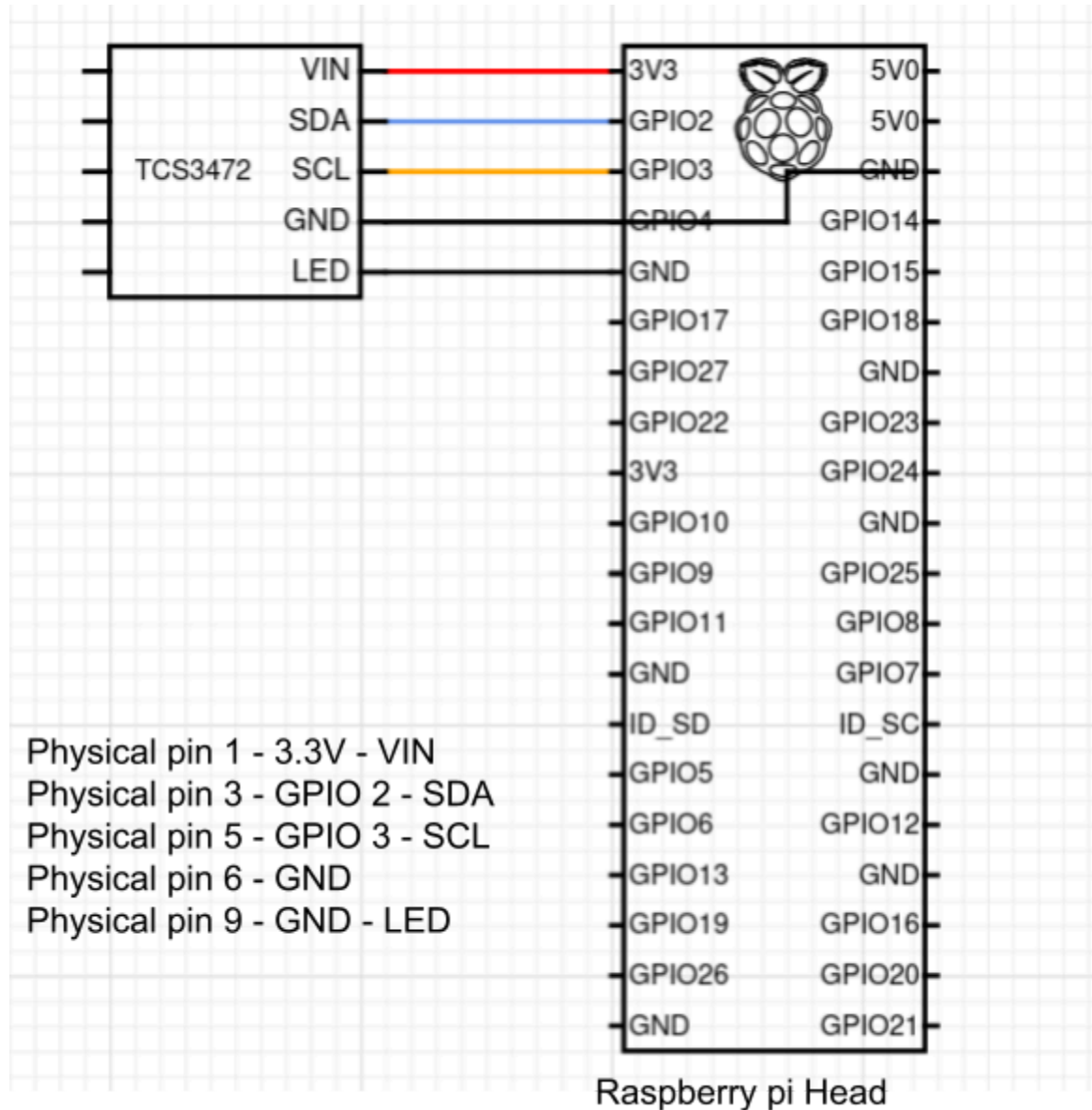
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Photo of completed circuit:



Hardware Diagram:



Screen shot of compilation:

```
pi40@raspberrypi40:~/EmbeddingLinux/csc-615-assignment-5-rgb-sensor-Jasuv $ make
mkdir -p dist
gcc -Wall -g -O2 -Ilib -c main.c -o dist/main.o
gcc -Wall -g -O2 -Ilib -c lib/ColorLib.c -o dist/ColorLib.o
gcc -Wall -g -O2 -Ilib -c lib/DEV_Config.c -o dist/DEV_Config.o
gcc -Wall -g -O2 -Ilib -o assignment5 dist/main.o dist/ColorLib.o dist/DEV_Config.o -lm -lbcm2835
Build Successful: Executable 'assignment5' created.
pi40@raspberrypi40:~/EmbeddingLinux/csc-615-assignment-5-rgb-sensor-Jasuv $
```

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Screen shot(s) of the execution of the program:

```
pi06@raspberrypi06:~/CSC615/csc-615-assignment-5-rgb-sensor-Jasuv $ make run
sudo ./assignment5
bcm2835 init success !!!
R:136 G:255 B:28
Best: Yellow dist=27.6 confidence=93.8%
Hex: #88FF1C | Color: Yellow
R:136 G:255 B:28
Best: Yellow dist=27.6 confidence=93.8%
Hex: #88FF1C | Color: Yellow
R:134 G:255 B:27
Best: Yellow dist=27.7 confidence=93.7%
Hex: #86FF1B | Color: Yellow
^C
Caught Ctrl+C (SIGINT), exiting program.
pi06@raspberrypi06:~/CSC615/csc-615-assignment-5-rgb-sensor-Jasuv $ ^C
```